

# Sequence Learning – Recurrent Neural Network - LSTM

# Sequence Learning Problems

- In feed-forward and convolutional neural networks the size of the input was always fixed.
- For example, we fed fixed size (32 x 32) images to convolutional neural networks for image classification.
- Each input to the network was independent of the previous or future inputs.
- The computations, outputs and decisions for two successive images are completely independent of each other.

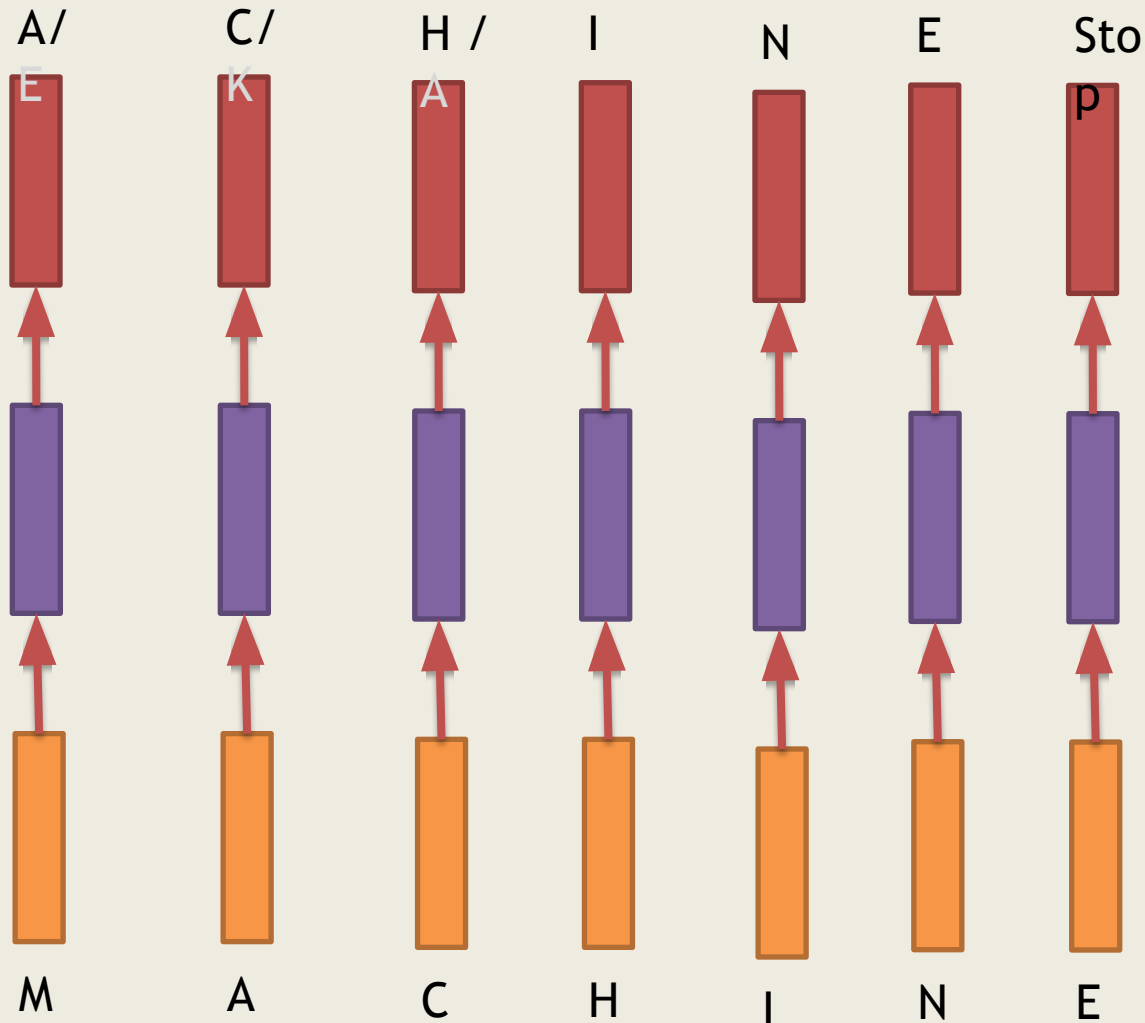
# Sequence Learning Problems

- In many applications the input is not of a fixed size.
- Successive inputs may not be independent of each other.
- The length of the inputs and the number of predictions you need to make is not fixed.



Task of  
autocompletion

# Sequence Learning Problems



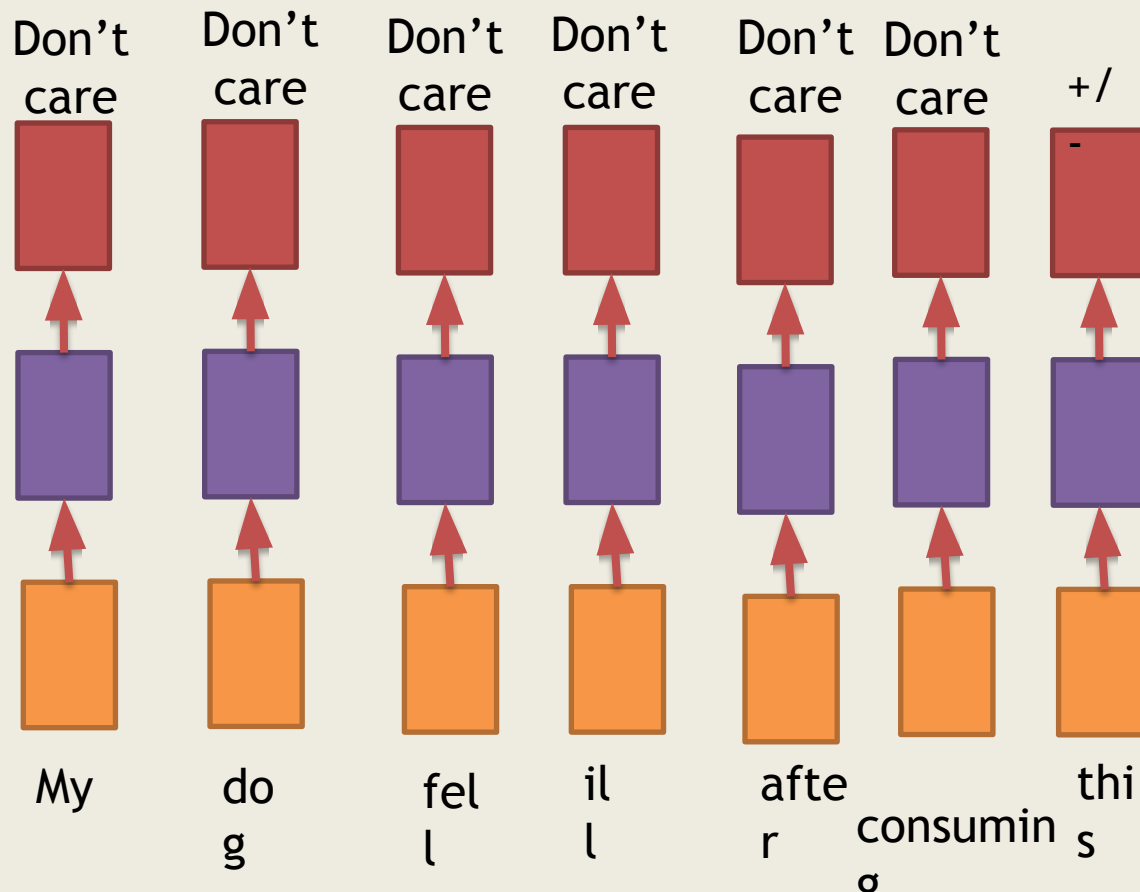
Each network is performing the task of input character output

# Sequence Learning Problems

- To look at a sequence of (dependent) inputs and produce an output (or outputs)
- Each input corresponds to one time step
- Sometimes we may not be interested in producing an output at every stage
- Instead we would look at the full sequence and then produce an output

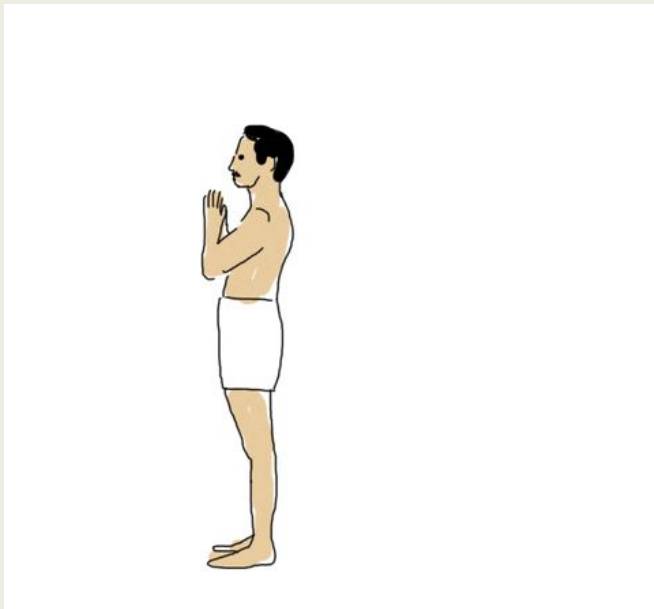
# Sequence Learning Problems

- Consider the task of predicting the polarity of a dog food.

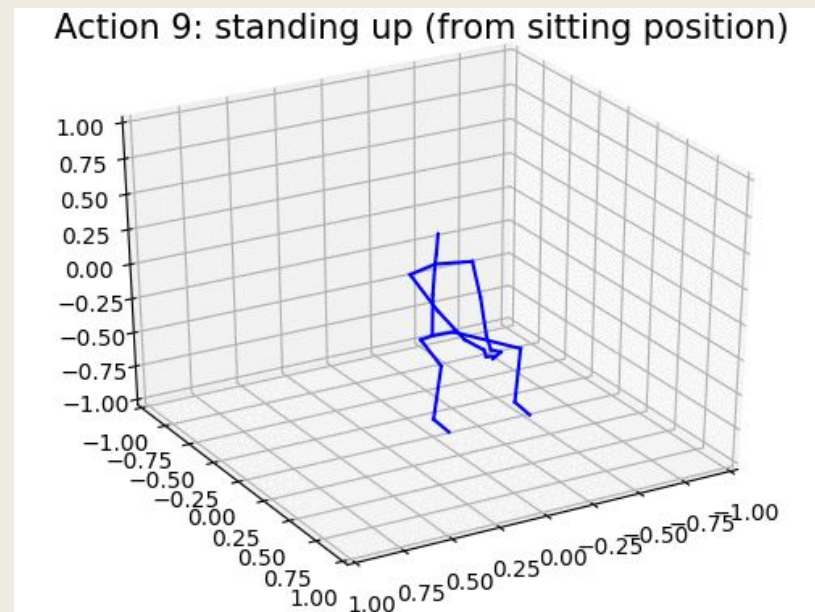


# Sequence Learning Problems

- Sequences could be composed of any thing (not just words)



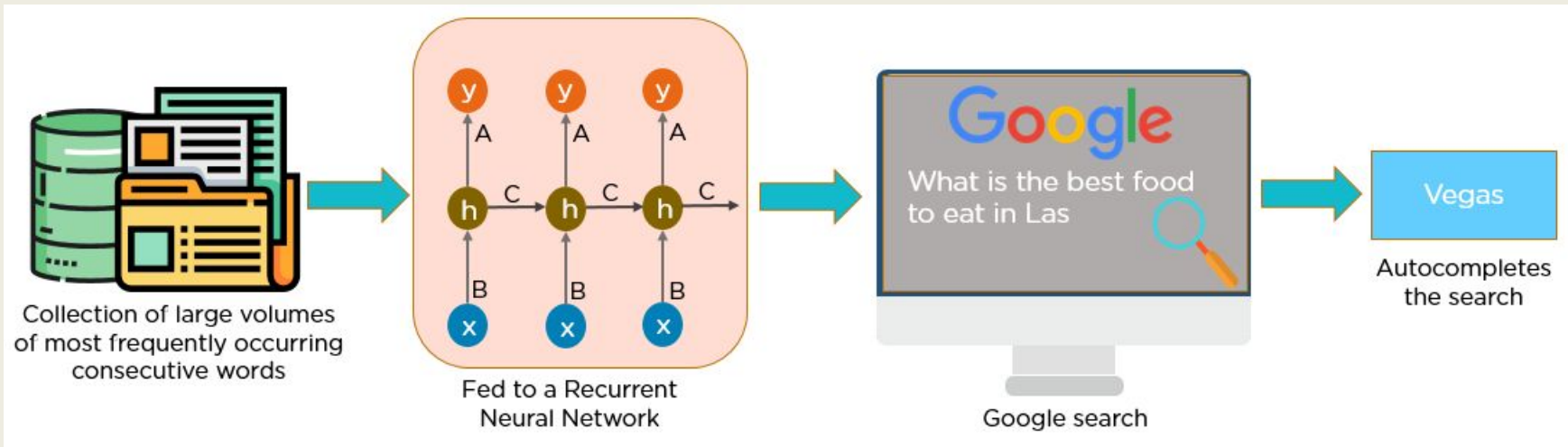
Surya  
Namaskar



Human Activity  
Recognition

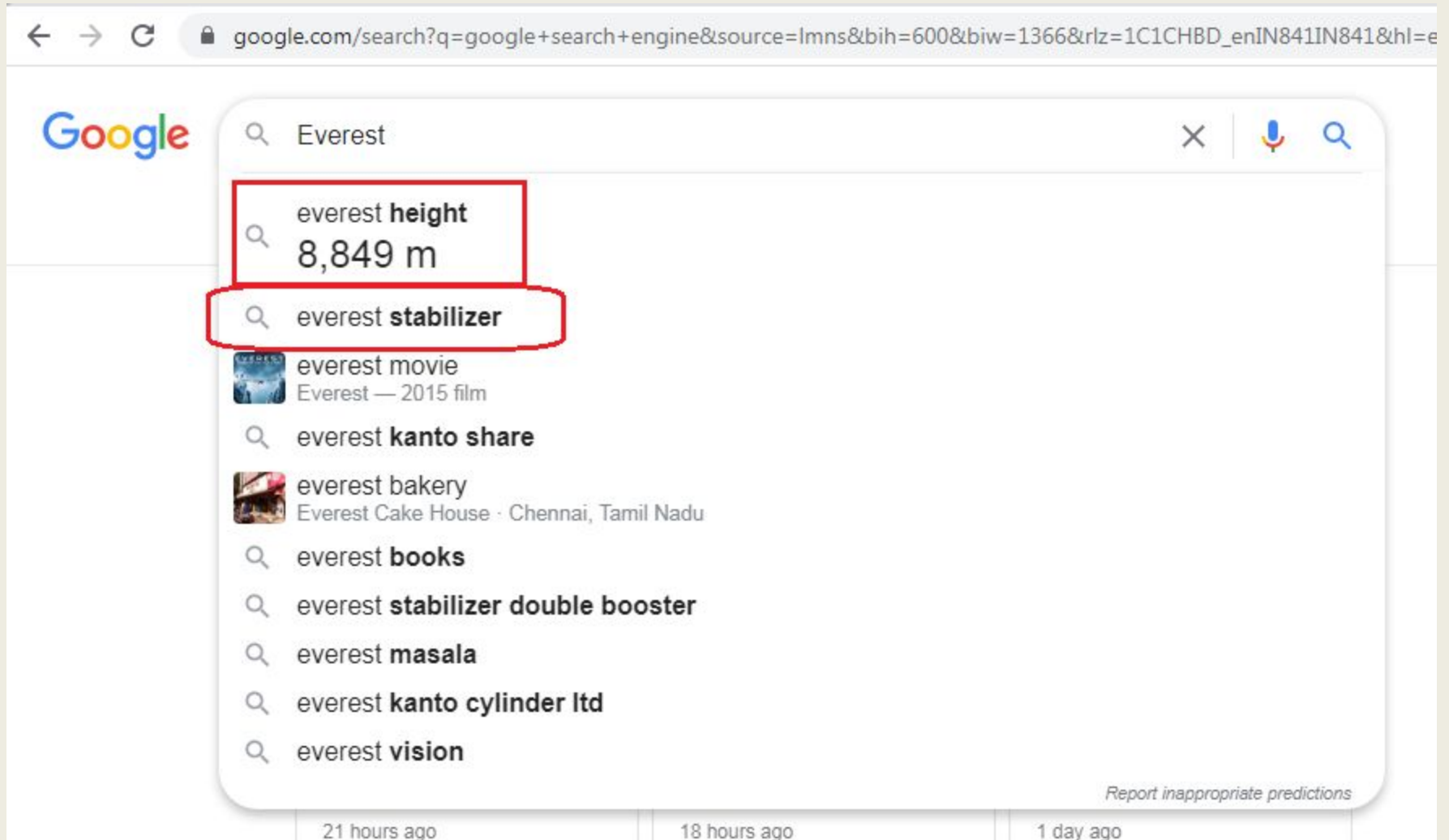
# Recurrent Neural Network

- A recurrent neural network (RNN) is a type of artificial neural network.
- Designed to recognize a data's sequential characteristics and use patterns to predict the next likely scenario.





# Recurrent Neural Network



A screenshot of a Google search page. The search bar contains the text "Everest". Below the search bar, a list of suggestions is displayed. The first suggestion is "everest height 8,849 m", which is highlighted with a red rectangular box. The second suggestion is "everest stabilizer", which is also highlighted with a red rectangular box. Other suggestions include "everest movie", "everest kanto share", "everest bakery", "everest books", "everest stabilizer double booster", "everest masala", "everest kanto cylinder ltd", and "everest vision". At the bottom right of the suggestions list, there is a link that says "Report inappropriate predictions".

← → ↻ google.com/search?q=google+search+engine&source=lmns&bih=600&biw=1366&rlz=1C1CHBD\_enIN841IN841&hl=e

Google

Everest

everest **height**  
8,849 m

everest **stabilizer**

everest movie  
Everest — 2015 film

everest **kanto share**

everest bakery  
Everest Cake House · Chennai, Tamil Nadu

everest **books**

everest **stabilizer double booster**

everest **masala**

everest **kanto cylinder ltd**

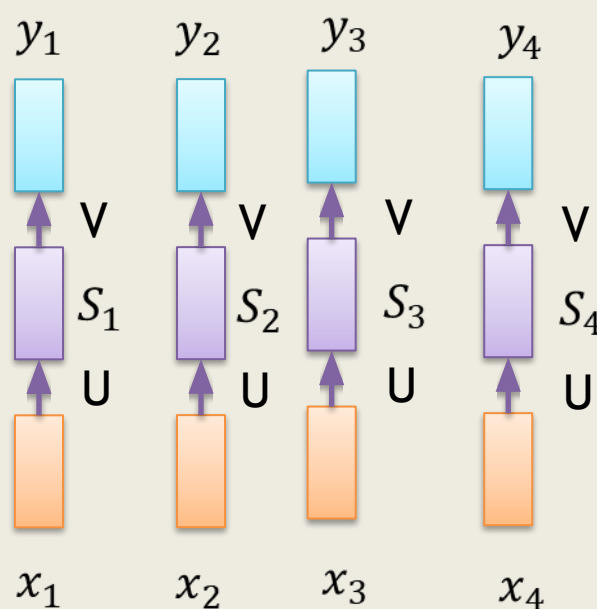
everest **vision**

Report inappropriate predictions

21 hours ago 18 hours ago 1 day ago

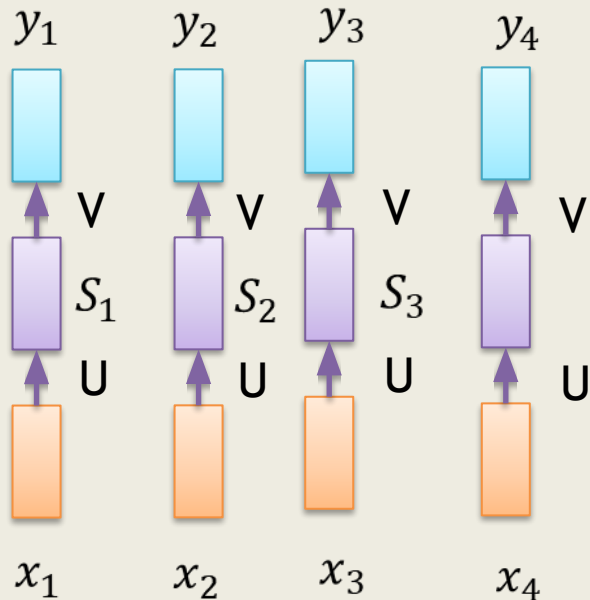
# Recurrent Neural Network

- Account for dependence between inputs
- Account for variable number of inputs
- Make sure that the function executed at each time step is the same



Since, we want the same function to be executed at each timestep we should share the same network (i.e. Since we are simply going to compute the same function (with same parameters) at each timestep, the number of timesteps doesn't matter

# Recurrent Neural Network



$x_t$  is the input to the network at time step  $t$

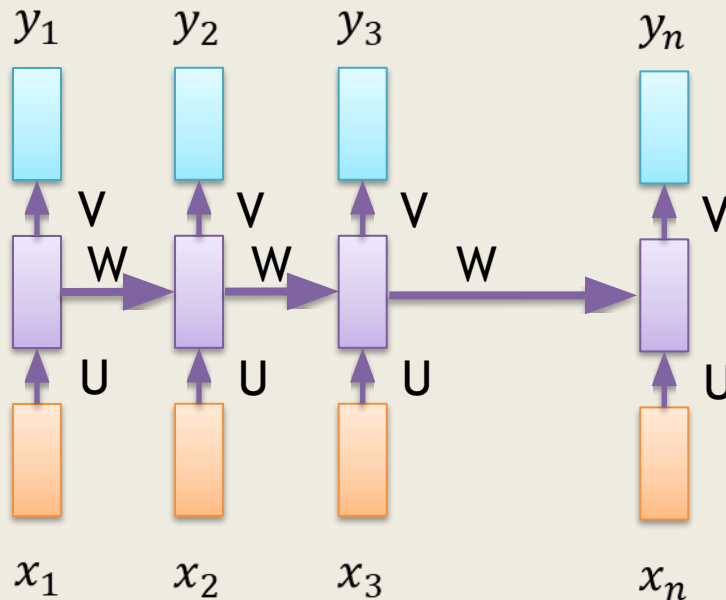
Example : one-hot vector corresponding to a word of a sentence.

**Weights** : parameterized by a weight matrix  $U$ , hidden-to-hidden recurrent connections parameterized by a weight matrix  $W$ , and hidden-to-output connections parameterized by a weight matrix  $V$  and all these weights ( $U, V, W$ ) are shared across time.

**Bias** :  $c$  and  $b$

# Recurrent Neural Network

- Account for dependence



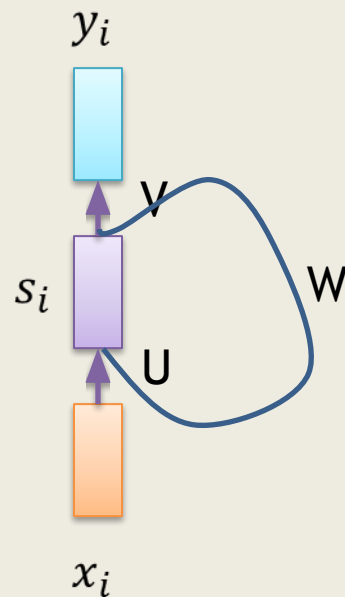
$$s_i = \sigma(Ux_i + Ws_{i-1} + b)$$

$$y_i = \mathcal{O}(Vs_i + c)$$

Where  $s_i$  is the state of the Network at timestep  $i$   
Parameters  $W, U, V, c, b$  are shared across timesteps

# Recurrent Neural Network

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Thank You